

LECTURE 28

Mass Storage Management

CSC323 · Operating Systems

COMSATS University Islamabad, Lahore Campus

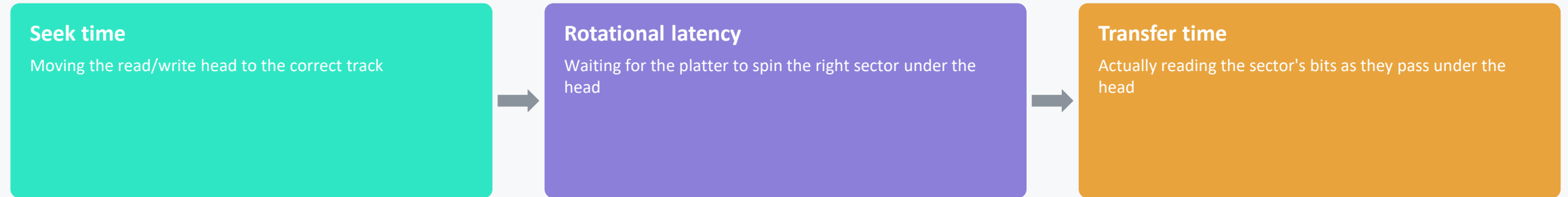
In This Lecture

- The physical structure of a hard disk, and the three components of disk access time
- **Volatile vs. non-volatile memory**, and how SSD performance characteristics differ from a mechanical disk's
- **Why disk scheduling matters**: seek time dominates, so request order matters
- **FCFS, SSTF, SCAN, and C-SCAN**, worked against one shared example
- **Why NVM/SSD scheduling** is a fundamentally different problem from head-movement minimization

Hard Disk Structure and Access Time

A hard disk stores data on spinning platters, each with its own read/write head. Each surface is divided into tracks, each track into sectors — the smallest unit the disk can read or write. Reading any sector requires three distinct delays, all happening in sequence.

The three components of hard-disk access time



All three must happen in sequence on every access. There is no way to skip or overlap seek, rotational latency, and transfer for a single mechanical read.

Worked Example: Disk Access Time

A disk spinning at 7,200 RPM, average seek time 4 ms, transfer rate 125 MB/s, reading a single 4 KB block.

Component	Calculation	Value
Seek time	given	4 ms
Rotational latency	$(60,000 / 7,200) \div 2 \approx 8.33 / 2$	≈ 4.17 ms
Transfer time	$4,096 \text{ bytes} / 125,000,000 \text{ bytes/s}$	≈ 0.033 ms

Total access time = 4 + 4.17 + 0.033 $\approx$ 8.2 ms. This is exactly the page-fault service time Lecture 25's EAT worked example used — a page fault's cost is dominated almost entirely by this one disk access, roughly forty thousand times slower than a 200 ns memory access.

Volatile and Non-Volatile Memory

RAM is volatile: its contents are lost the instant power is removed. Flash memory, SSDs, and other non-volatile memory (NVM) retain their contents with no power applied at all.

- **No moving parts.** An SSD pays neither seek time nor rotational latency — any block is reached in roughly the same, very small time, regardless of where it sits.
- **Limited write-erase cycles.** Flash cells wear out after a limited number of write-erase cycles, trading the mechanical bottleneck for a different one.
- **Wear leveling.** SSD controllers spread writes evenly across all cells, so no one region wears out and fails long before the rest of the drive.

Disk Scheduling

For a mechanical hard disk, seek time dominates total access time — and it's the one component that depends heavily on where the head currently is versus where it needs to go next. This makes the order pending requests are serviced matter enormously.

- **FCFS** (First-Come, First-Served) — service requests in arrival order. Simple, but makes no attempt to minimize head movement.
- **SSTF** (Shortest Seek Time First) — always service whichever request is closest to the head. Minimizes movement greedily, but can starve far-away requests.
- **SCAN** (the “elevator algorithm”) — sweep in one direction, servicing every request along the way, then reverse.
- **C-SCAN** — sweeps one direction only; jumps back to the start without servicing, then sweeps forward again, trading movement for uniform wait times.

FCFS: Service in Arrival Order

Request queue, in arrival order: 98, 183, 37, 122, 14, 124, 65, 67.

Head position sequence — 200 cylinders (0–199), starting at cylinder 53

53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67

Move distances

Move	Distance
53 → 98	45
98 → 183	85
183 → 37	146
37 → 122	85
122 → 14	108
14 → 124	110
124 → 65	59
65 → 67	2

Total FCFS movement: 640 cylinders.

SSTF: Always Jump to the Closest Request

Same queue, same starting position — always service whichever pending request is closest to the head.

Head position sequence — 200 cylinders (0–199), starting at cylinder 53

53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183

Move distances

Move	Distance
53 → 65	12
65 → 67	2
67 → 37	30
37 → 14	23
14 → 98	84
98 → 122	24
122 → 124	2
124 → 183	59

Total SSTF movement: 236 cylinders — dramatically less than FCFS, for the identical requests.

SCAN: Sweep to the End, Then Reverse

Sweeping upward (toward higher cylinder numbers) first, all the way to the end of the disk, then reversing.

Head position sequence — 200 cylinders (0–199), starting at cylinder 53

53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 37 → 14

Move distances

Move	Distance
53 → 65	12
65 → 67	2
67 → 98	31
98 → 122	24
122 → 124	2
124 → 183	59
183 → 199 (end of disk)	16
199 → 37	162
37 → 14	23

Total SCAN movement: 331 cylinders.

C-SCAN: One-Direction Sweep With a Return Jump

Sweeping upward only; jumping back to cylinder 0 without servicing on the return, then continuing upward.

Head position sequence — 200 cylinders (0–199), starting at cylinder 53

53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 0 → 14 → 37

Move distances

Move	Distance
53 → 65	12
65 → 67	2
67 → 98	31
98 → 122	24
122 → 124	2
124 → 183	59
183 → 199 (end of disk)	16
199 → 0 (return jump, no service)	199
0 → 14	14
14 → 37	23

Total C-SCAN movement: 382 cylinders.

Head Movement Comparison

Same request queue, same starting position (cylinder 53)

Algorithm	Total head movement (cylinders)
FCFS	640
SCAN	331
C-SCAN	382
SSTF	236 (lowest)

SSTF wins on raw total movement, as it usually does. But C-SCAN moves more than SCAN overall, despite being the one specifically designed for fairness — it spends extra movement on its unserviced return jump to guarantee every request waits roughly the same amount of time, rather than to minimize total movement.

SSTF's starvation risk doesn't show up in a single snapshot. A steady stream of requests near the head's current position can starve a lone far-away request indefinitely — something closer always wins the “shortest seek” comparison.

NVM Scheduling

None of the four algorithms above solve a seek-time problem for flash-based storage, because flash has no seek time to minimize in the first place. NVM scheduling is a genuinely different problem.

- **Write amplification** — a single logical write can force the controller to rewrite a much larger physical region, because flash can only be erased in large blocks.
- **Parallelism across flash channels** — an SSD is typically built from many independent flash chips wired to separate channels, so scheduling can dispatch multiple requests simultaneously.

The mental model shifts entirely. A hard-disk scheduler's job is minimizing where the head has to go next; an NVM scheduler's job is minimizing wear and maximizing how much work happens in parallel — the same high-level goal, solved against a completely different set of physical constraints.

Key Takeaways

- **Hard-disk access time** has three components — seek time, rotational latency, transfer time — combining to roughly 8.2 ms for a typical 7,200 RPM disk.
- **RAM is volatile; flash/SSD/NVM is non-volatile.** SSDs pay no seek or rotational cost, but manage limited write-erase cycles through wear leveling.
- **Request order matters enormously:** the same eight-request queue cost 640 cylinders under FCFS, but only 236 under SSTF, with SCAN (331) and C-SCAN (382) trading movement for fairness.
- **SSTF** minimizes movement greedily but can starve far-away requests; SCAN and C-SCAN bound worst-case waiting by sweeping systematically.
- **NVM/SSD scheduling** is a different problem altogether — write amplification and parallelism across flash channels, not head movement.